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Impact of Fluid Exposure History on Drill Cuttings Mechanical Properties Through Indentation Testing

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Abstract

Understanding the mechanical properties of rocks throughout the full length of a well is critical for optimizing drilling and production operations. Indentation testing offers a low-cost and time-effective method to obtain geomechanical properties from cuttings and enables access to data that were previously unattainable. Rock cuttings are a widely abundant resource of information, but encounter various fluid exposure histories before collection, which can alter their properties through hydro-chemo-mechanical interactions. This study investigates the influence of fluid exposure on the mechanical properties of cuttings using indentation testing.

Four rock types with distinct mineralogies (Eagle Ford Shale, Mancos Shale, Indiana Limestone, and Crab Orchard Sandstone) were tested. Fluid exposure was simulated in two stages: a 24-hour soak in base fluids to represent formation fluids (oil-bearing or water-bearing), followed by a 1-hour soak in drilling fluids (water-based or oil-based mud). Four combinations of fluid exposure were tested accordingly for each rock type: oil-oil, oil-water, water-water, and water-oil. Indentation tests measured hardness and reduced modulus, with 16 indentations performed prior to fluid exposure and 8 each on soaked samples before and after polishing.

Fluid history had little effect on Eagle Ford (±6 % modulus and hardness) and on the scatter-dominated Crab Orchard sandstone. Mancos shale stiffened when kept in oil (+26 % modulus, +35 % hardness) but weakened severely after an oil-to-water reversal (–42 % modulus, –48 % hardness), confirming the importance of wetting order. Indiana limestone behaved oppositely: any path ending in oil raised stiffness, whereas continuous water reduced modulus by 20 % and halved hardness, evidence of rapid carbonate dissolution.

The results show that fluid sequence can shift indentation properties from negligible change to nearly fiftypercent loss. Recording oil–water contact history, protecting shale and limestone cuttings from prolonged water, and applying statistical treatment to coarse sandstones are practical steps to improve cuttings-based geomechanical profiles.

Introduction

Rock-mechanical data control many drilling and production decisions. Elastic modulus and rock strength inform wellbore-stability models, mud-weight windows, completion design, and post-drilling geomechanics. Operators usually obtain these parameters from sonic or density logs and from triaxial testing of whole-core plugs. Both routes have cost and resolution limits. Logs average properties over decimeters and use empirical correlations to calculate the mechanical properties. Core plugs are scarce, expensive to obtain and test, and are only available for reservoir sections. The result is a patchy mechanical profile along the well, which forces engineers to rely on broad safety factors.

Instrumented indentation on drill cuttings offers a practical alternative. Cuttings are produced continuously while drilling and already reach the surface for routine lithological description. A single indentation measurement gives indentation modulus and indentation hardness. Repeated indents across a cutting supply millimeter-scale resolution at a fraction of the cost and time of traditional tests (Gramajo et al., 2024). Recent studies have correlated indentation modulus with static Young's modulus for carbonates, sandstones, and shales, demonstrating that cuttings can capture small-scale heterogeneity missed by logs and core plugs (Abdallah et al., 2023; Chen et al., 2015; Mousavi et al., 2018).

Fluid exposure history remains as a major uncertainty. As cuttings travel to the surface and before mud-logging crews collect samples, cuttings reside in the borehole for minutes to hours, immersed first in formation fluids and then in drilling mud. Capillary imbibition, dissolution–precipitation reactions, and clay swelling can modify the microstructure and, by extension, the measured mechanical response. Reports on shale–fluid chemo-mechanics show modulus reductions of up to 20 % depending on water saturation, yet the data sets focus on bulk compression tests or acoustic velocity, not instrumented indentation (Liu et al., 2018; Wong et al., 2016). For sandstones and carbonates, the literature is even more limited, and no cross-lithology comparison under a common protocol exists. Without quantifying these effects, uncertainty in cuttings-based geomechanics persists and may lead to errors in interpreting indentation measurements.

The present work addresses this gap through a systematic indentation study on four rocks with contrasting mineralogy: Eagle Ford shale (calcite–quartz), Mancos shale (quartz-rich with minor carbonates and clays), Indiana limestone (calcite), and Crab Orchard sandstone (quartz). Each sample underwent a two-stage exposure sequence that represents typical downhole history: a 24 h soak in the resident formation fluid followed by a 1 h soak in the active drilling mud. Two fluids were selected to isolate wettability and chemical effects. Additive-free mineral oil was selected as a non-polar medium and local tap water as a brine proxy. Four histories resulted: oil→oil (OO), oil→water (OW), water→water (WW), and water→oil (WO). Indentation tests at a peak load of 7.5 N were performed on the dry state and after each soak, allowing direct measurement of property changes.

The objectives of the experimental procedure are to 1) quantify the magnitude and direction of change for indentation parameters for each rock–fluid combination, 2) relate the observed trends to mineral composition and the order of fluid contact, and 3) propose practical guidelines for handling, preparing, and interpreting drill-cutting indentations so that downhole geomechanical models reflect in-situ conditions rather than artefacts of surface exposure.

Materials and methods

Rock samples

Four sedimentary rocks that represent shale, carbonate and sandstone lithologies were investigated: Eagle Ford shale, Mancos shale, Indiana limestone and Crab Orchard sandstone. Samples measuring ~7 mm × 10 mm × 10 mm were cut from cores (Fig. 1). A final thickness of at least 10 mm eliminates substrate influence on the measured response. All faces were sequentially polished with silicon-carbide papers to P4000 (Rached et al., 2024). Table 1 summarizes the bulk mineral composition of the samples obtained from powder X-ray. Eagle Ford consists of 76 wt % calcite and 24 wt % quartz; Mancos contains 85wt % quartz, 7 wt % dolomite and 5 wt % calcite with <3% illite–smectite mixed layer clays; Indiana limestone is 100 wt% calcite, and Crab Orchard sandstone is 100 wt% quartz. Fig. 2 shows pre-exposure scanning electron images for the four samples.

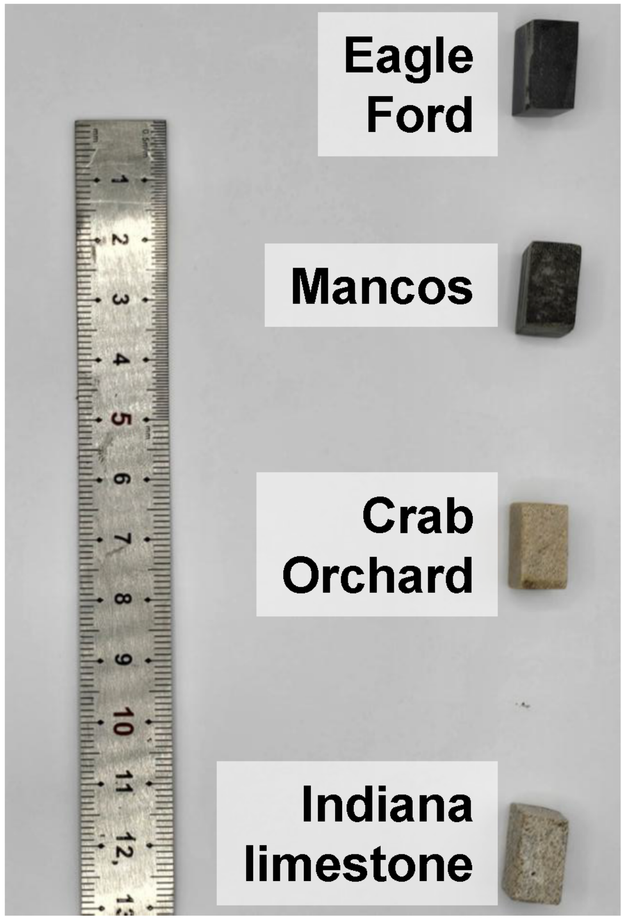

Figure 1—Picture of typical samples prepared for the exposure tests.

Table 1—Bulk mineral composition of the tested rock samples.

Rock	Main mineral constituents (wt %)
Eagle Ford shale	76 calcite, 24 quartz
Mancos shale	85 quartz, 7 dolomite, 5 calcite, <3 clays
Indiana limestone	≈ 100 calcite
Crab Orchard sandstone	≈ 100 quartz

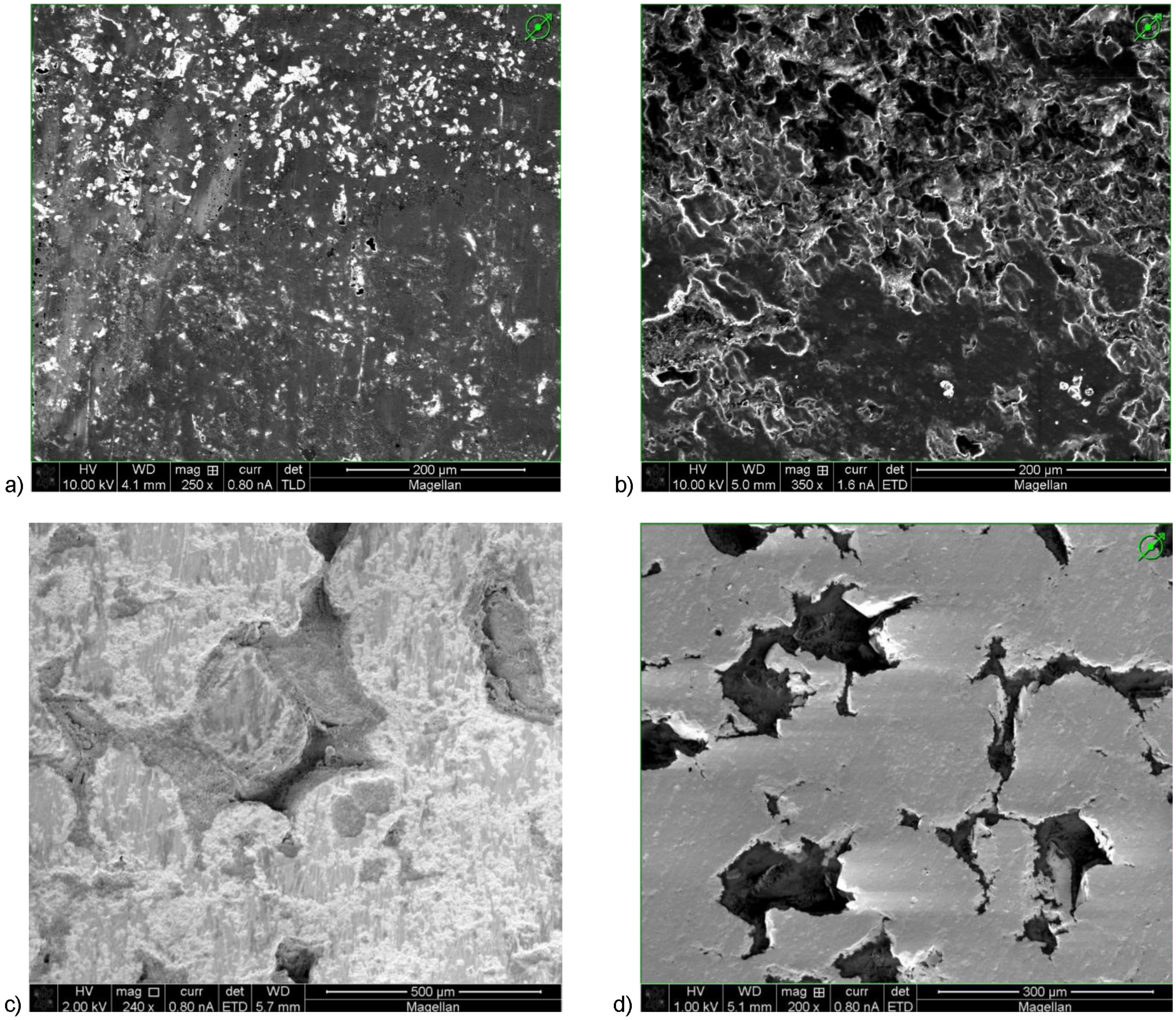

Figure 2—Pre-exposure scanning electron images of the four samples: a) Eagle Ford shale, b) Mancos shale, c) Indiana limestone, d) Crab Orchard sandstone

Fluids

Two simple fluids reproduced the downhole sequence of resident formation liquid followed by drilling mud. A non-polar additive-free white mineral oil (viscosity grade ISO VG 10 and density 0.85 g cm^{-3} at 25 °C) represented an oil-bearing formation. Local tap water with pH 7.6 and total dissolved solids near 0.5 g L^{-1}, served as a brine proxy and as the basis for a water-based mud. All exposures and mechanical tests were carried out at 23 ± 1 °C.

Fluid-exposure protocol

Each sample was first placed under a mild vacuum for 30 min while submerged in the designated first fluid to remove near-surface air and to accelerate saturation. The sample then soaked statically in that fluid for 24 h. The sample was then immediately immersed in the second fluid for an additional 1 h, reflecting the short residence time of cuttings in active drilling mud as the flow back to the surface and at shale shakers. Four exposure histories were generated: oil followed by oil (OO), oil followed by water (OW), water followed by water (WW) and water followed by oil (WO). Control specimens were stored oven-dry at 60 °C.

Sample preparation before indentation

Immediately before indentation testing, working faces were inspected under reflected light. When required, the surface was repolished by a brief 30 s grind on P4000 paper to remove any adsorbed films. A square grid was lightly scribed to space indentations apart so that individual plastic zones did not overlap.

Indentation testing

Instrumented indentations were performed with an Anton Paar MCT-3 tester equipped with a Vickers diamond indenter. The load function comprised a 30 s ramp from 0.05 N to the peak load of 7.5 N, a 10 s dwell at peak to reduce visco-elastic creep effects and a 30 s unload back to 0.05 N. Instrument compliance and thermal drift were corrected automatically using a post-test baseline segment. Sixteen indents were completed on the dry cube of each rock. After the fluid-soak sequence eight indents were conducted on the unpolished surface and a further eight on the repolished surface. Tests that exhibited pop-in events, incomplete unloading or edge contact were rejected. The full testing workflow is illustrated in Fig. 3.

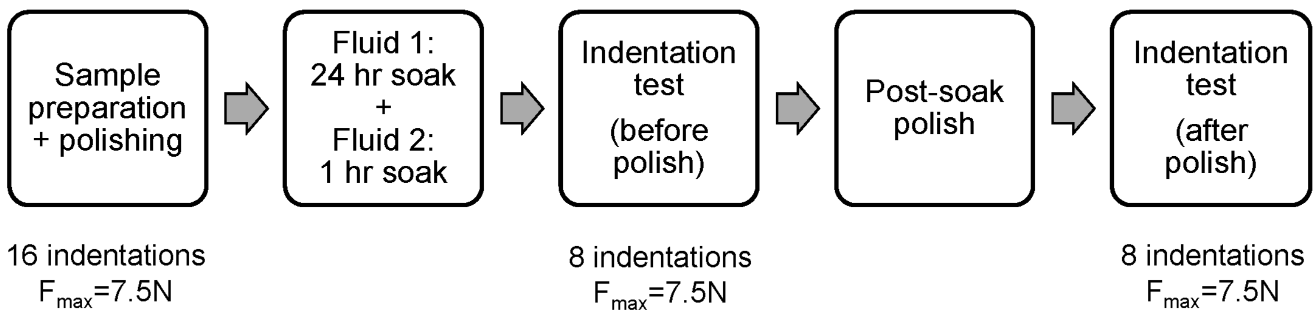

Figure 3—Sample preparation, fluid exposure, and testing procedure workflow.

Data analysis

Load–displacement curves were analyzed with the Oliver–Pharr method to obtain the reduced modulus (Er). The projected contact area was calculated from the indenter order area function. A Poisson's ratio of 0.3 was assumed for all rocks. Indentation hardness (HIT) was defined as peak load divided by contact area. Individual results lying more than 1.5 times the inter-quartile range outside the first or third quartile were treated as statistical outliers and removed. For every rock–fluid condition at least twelve valid indents remained, from which the mean, standard deviation and coefficient of variation were calculated. Linear regressions of Er against HIT for each exposure path were also fitted to evaluate consistency among indents and to identify any fluid-dependent trends.

Results

A total of 320 valid load–displacement curves were analyzed: sixteen indents on the dry reference of each rock and sixty-four after fluid exposure (eight before polish and eight after polish for each of the four histories). Figures 4–7 plot reduced modulus Er versus indentation hardness HIT for Eagle Ford shale, Mancos shale, Indiana limestone and Crab Orchard sandstone, respectively, with linearleast-squares trends super-imposed.

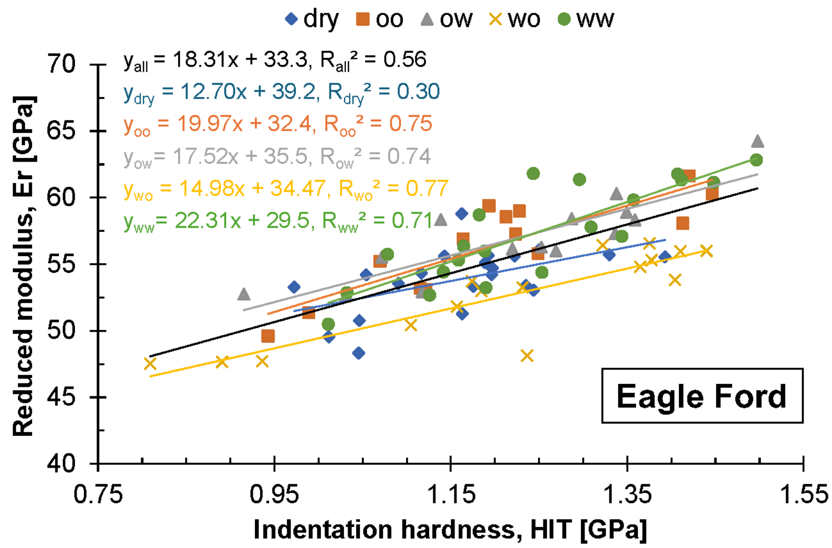

Figure 4—Reduced modulus versus indentation hardness for Eagle Ford samples for different fluid exposure histories with linear fits. Blue diamonds: dry, orange squares: oil → oil, gray rectangles: oil → water, yellow "x": water → oil, green circle: water→water.

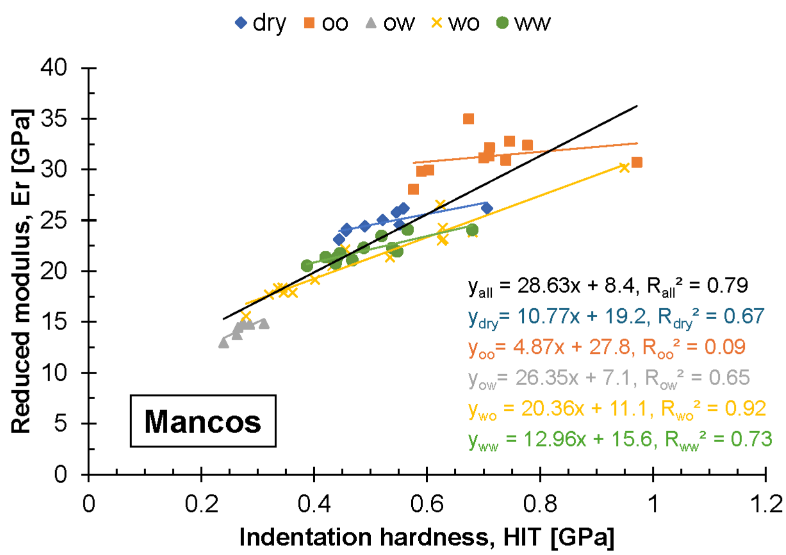

Figure 5—Reduced modulus versus indentation hardness for Mancos shale samples for different fluid exposure histories with linear fits. Blue diamonds: dry, orange squares: oil → oil, gray rectangles: oil → water, yellow "x": water → oil, green circle: water→water.

Figure 6—Damaged Mancos shale chip following oil → water exposure.

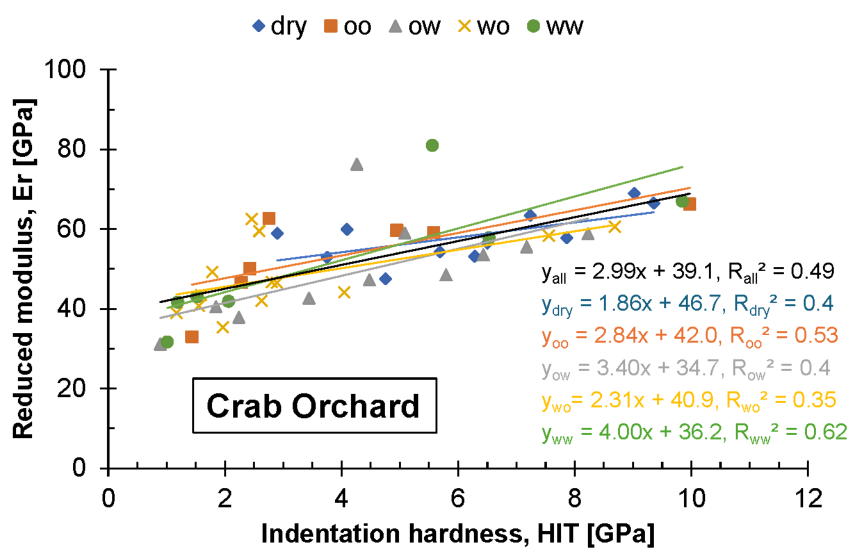

Figure 7—Reduced modulus versus indentation hardness for Indiana limestone samples for different fluid exposure histories with linear fits. Blue diamonds: dry, orange squares: oil → oil, gray rectangles: oil → water, yellow "x": water → oil, green circle: water→water.

Eagle Ford shale

Dry Eagle Ford samples cluster around Er ≈ 54 GPa and HIT ≈ 1.1 GPa. None of the four exposure paths produces a systematic shift (Fig. 4). Mean changes remain within ±6% for both properties, and regression lines overlap the dry trend. The absence of a fluid effect agrees with the low clay content and very fine grain size, which minimize capillary uptake and surface reactions during the relatively short one-day exposure.

Mancos shale

Mancos shows the strongest response (Fig. 5). The oil→oil sequence raises Erby 26% and HIT by 35%, indicating stiffening and hardening. In contrast, water→water and water→oil substitutions reduce modulus by ~15% and hardness by ~5%. Oil→water causes severe softening, with Er decreasing by 42% and HIT by 48%. Several samples subjected to oil→water crumbled during post-soak polishing, confirming significant weakening (Fig. 6). Because illite–smectite mixed layers are below 3%, classical clay swelling is unlikely. Results point to other chemical interactions and changes in capillary stress when a wetting reversal occurs.

Indiana limestone

Indiana limestone responds noticeably to fluid history (Fig. 7). Oil followed by oil had a negligible effect. Oil followed by water gives larger gains, 10 % in Er and 12 % in HIT, while water followed by oil yields roughly 7 % increases in both parameters. Continuous water exposure has the opposite effect, reducing the modulus by 20 % and halving the hardness. These trends suggest that even brief oil contact preserved or stiffened the calcite framework, whereas prolonged water residence dissolves cement and weakens grain contacts, although absolute values still fall within the typical range for dense limestones.

Crab Orchard sandstone

Crab Orchard data display the widest scatter (Fig.8). Individual points span Er from about 32 to 80 GPa. This dispersion reflects the indentation impression size approaching the mean quartz-grain diameter, so local grain support rather than fluid exposure dominates compliance. Statistical averages differ little among the exposure histories, yet the poor repeatability restricts firm conclusions. These results emphasize that coarse sandstones may require higher loads or statistical corrections when using indentation for property estimation.

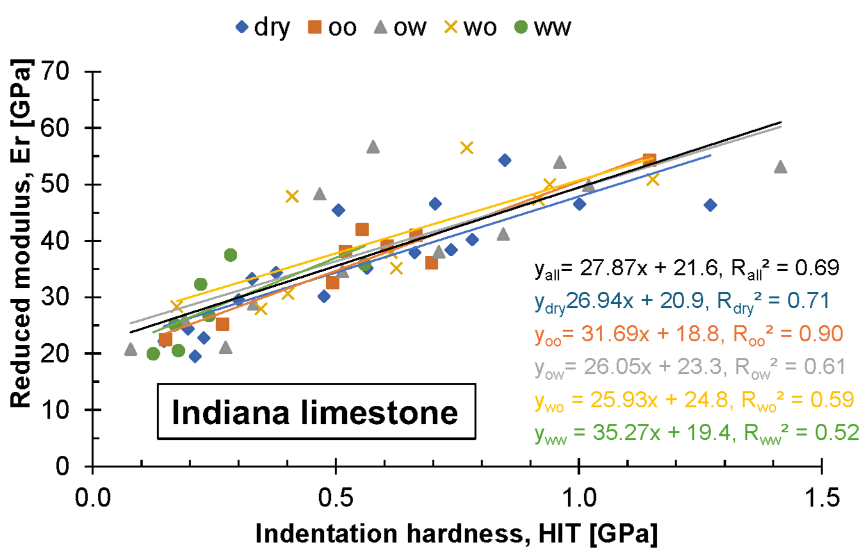

Figure 8—Reduced modulus versus indentation hardness for Crab Orchard sandstone samples for different fluid exposure histories with linear fits. Blue diamonds: dry, orange squares: oil → oil, gray rectangles: oil → water, yellow "x": water → oil, green circle: wate→water.

Relationship between Erand HIT

Across all rocks and conditions, Er and HIT maintain a near-linear correlation. Coefficients of determination exceed 0.70 for Eagle Ford (OO, OW, WO, WW), Indiana limestone (dry, OO), and Mancos (WO, WW), confirming that hardness can serve as a quick proxy for modulus. Crab Orchard samples show lower correlations due to the scatter in the data, with R^2<0.62.

Summary of fluid effects

Table 2 summarizes the percentage changes in the reduced modulus relative to the dry state. Fluid exposure can be ignored for the Eagle Ford samples and Crab Orchard samples over a one-day time frame, whereas the Mancos shale requires explicit correction for the saturation history. In practical drilling workflows, capturing the order of fluid contact and minimizing water substitution on shale cuttings will improve the reliability of cuttings-based geomechanics.

Table 2—Changes in the reduced modulus (Er) for the four rock samples under four exposure histories.

Rock	OO	OW	WW	WO
Eagle Ford	+4%	+6%	+6%	-3%
Mancos	+26%	−42 %	−11%	−15%
Indiana	+3%	+10%	-20%	+15 %
Crab Orchard	±scatter, no clear trend			

Discussion

The fluid-exposure experiments reveal that the mechanical response of drill cuttings is governed by a combination of mineralogy and the sequence in which oil and water contact the rock. Although 24 h represents only a brief segment of downhole residence, the results already span from negligible change in fine-grained calcite–quartz shale to near one-half loss of hardness in limestone left in continuous water. Understanding the controls behind this spread is essential for reliable indentation measurements.

The small shifts recorded for Eagle Ford shale and the chemically neutral trend observed within the scatter of Crab Orchard sandstone confirm that rocks with limited reactive cement and low clay content are only weakly affected by the oil–water cycle tested. Eagle Ford cuttings showed modulus and hardness variations within 6% regardless of the exposure path. Such stability is attributed to a dense texture of calcite and microcrystalline quartz, which limits pore connectivity and shields grain contacts from dissolution or capillary stress changes. In Crab Orchard sandstone the indentation size approaches the grain diameter, and the >50% spread in elastic modulus across individual indents is dictated by whether the tip lands on a single quartz grain or a grain boundary. This geometric scatter masks any chemistry-driven change. These two lithologies therefore set useful bounds: cuttings from fine calcite–quartz shales can be indented without correction, while coarse sandstones need either larger loads or statistical averaging before fluid effects can even be resolved.

For Indiana limestone, when oil occupies the pore space at any point in the sequence, modulus and hardness increase. However, the changes in values remain <10% and within expected experimental variation. Continuous water exposure eliminates this protective phase, lowering modulus by 20% and halving hardness. Because all tests were at room temperature, these changes likely stem from preferential dissolution along microcracks or grain boundaries rather than bulk calcite solubility, yet the magnitude proves that mechanical weakening can develop within drilling time scales if limestone cuttings are left exposed to water-based mud.

Mancos shale shows the greatest sensitivity. Oil retained throughout the sequence produces a stiffening of 26% in modulus and 35% in hardness. Replacing oil with water triggers severe degradation: the oil-to-water path cuts modulus by 42% and hardness by 48%, and several samples crumble during polishing. Even without a wetting reversal, water followed by water still reduces modulus by 11% and water followed by oil by 15%. With illite–smectite content below 3%, classical clay swelling cannot account for these losses. Instead, capillary suction changes and dissolution of minor carbonate cement are potential drivers. In a quartz framework any weakening of cement quickly lowers contact stiffness, explaining the sharp decline in indentation properties.

Across all lithologies the order of fluid contact shows a clear influence. Sequences that finish in oil either preserve or enhance stiffness, whereas those ending in water generally soften the rock unless the initial oil phase leaves a protective film behind. The effect is most pronounced in carbonate-bearing compositions where aqueous dissolution is rapid. The presence or absence of a wetting reversal therefore deserves explicit recording in mud-logging workflows; neglecting it introduces up to 50% error in hardness for susceptible lithologies.

Several limitations bound the scope of the conclusions. Exposures were static and conducted at 23 °C; downhole circulation, elevated temperature and mud additives are expected to accelerate chemical reactions and may increase or decrease the trends reported. Only one mineral-oil grade and one tap-water chemistry were employed; salinity and pH variations could alter dissolution rates and capillary forces. Indentation depths sampled volumes on the order of cubic microns; in coarse sandstones this scale is smaller than a single grain, which is why scatter overwhelms chemistry. Finally, the study focused on short time frames relevant to cuttings retrieval; longer exposure could amplify the observed patterns. Future work should test dynamic flow conditions at elevated temperature, vary salinity and pH systematically, and employ higher loads or complementary micro-scratch methods for coarse-grained sandstones. Even within the present scope, the data provide immediate guidance: shale cuttings should be kept in oil-based mud or dried promptly, limestone cuttings require oil contact to limit dissolution, and sandstone cuttings demand statistical treatment rather than simple point measurements.

Conclusions

Indentation measurements on drill cuttings are reliable only when fluid-exposure history is understood. Fine-grained Eagle Ford shale and pure quartz Crab Orchard sandstone retain modulus and hardness within 6% after 24 h of oil- and water-soak sequences, so no correction is required for these lithologies. Mancos shale reacts strongly to wetting reversal: substituting oil with water lowers modulus by 42% and hardness by 48%, whereas continuous oil contact increases both properties. Pure-calcite Indiana limestone stiffens when oil is present at any stage but loses 20% of modulus and half of its hardness after uninterrupted water exposure, confirming rapid aqueous weakening of carbonate grain contacts. Across all rocks, sequences that finish in oil either preserve or enhance stiffness, while those ending in water soften the material unless an initial oil film remains. Recording the order of formation-fluid and mud contact, storing shale and limestone cuttings in oil-based mud or drying them quickly, and applying higher loads or statistical averaging on coarse sandstones will improve the accuracy of cuttings-based geomechanical profiles used in wellbore-stability modelling and drilling-fluid optimization.

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